# Algebra 2 Review: Linear Functions & Systems

**Station 1 of 4:** Equations of Lines, Slope, Parallel and Perpendicular

Name: _____ Date: _____

*Remember: perpendicular slopes are negative reciprocals, and $y - y_1 = m(x - x_1)$ has a minus sign in front of $y_1$.*

**1.** Write the equation of the line through $(-3, 5)$ and $(4, -9)$ in slope-intercept form.

**2.** The table shows a linear function $f$. Write $f(x)$ and find $f(20)$.

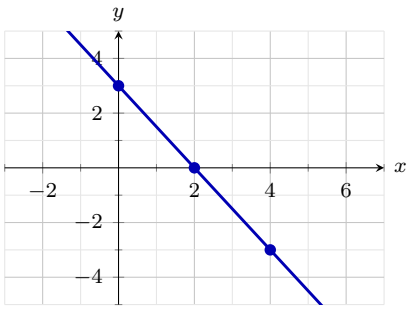
| $x$ | 2 | 5 | 8 | 11 |
|---|---|---|---|---|
| $f(x)$ | 7 | 13 | 19 | 25 |

**3.** A line has slope $-\frac{3}{4}$ and passes through $(8, -2)$. Write its equation in point-slope form and in slope-intercept form.

**4.** Write the line with slope $\frac{5}{2}$ through $(2, -3)$ in standard form $Ax + By = C$ with integer coefficients and $A > 0$.

**5.** Line $L$ is $2x + 5y = 10$. Write the equations of the lines through $(5, -1)$ that are (a) parallel to $L$ and (b) perpendicular to $L$.

**6.** Write an equation for the line shown and state its $x$-intercept.




**7.** Lines: $\ell_1:\ y = 2x - 5$, $\ell_2:\ 4x - 2y = 7$, $\ell_3:\ x + 2y = 4$. For each pair, state whether the lines are parallel, perpendicular, or neither.

**8.** A gym charges a \$45 joining fee plus \$28 per month. Write the total cost $C(m)$ for $m$ months, find $C(18)$, and find how many months cost \$605.

**9.** The line through $(k, 4)$ and $(2, -5)$ is perpendicular to $y = \frac{1}{3}x - 7$. Find $k$.

**10.** A data plan costs \$30 for up to 5 GB, plus \$8 for each additional GB. Write a piecewise function $C(g)$ and find $C(9)$.

## Algebra 2 Review: Linear Functions & Systems

**Station 2 of 4:** Linear Models, Scatter Plots, and Best-Fit Lines

Name: _____ Date: _____

*Interpret slopes with units, and do not extend a model far beyond the data it came from.*

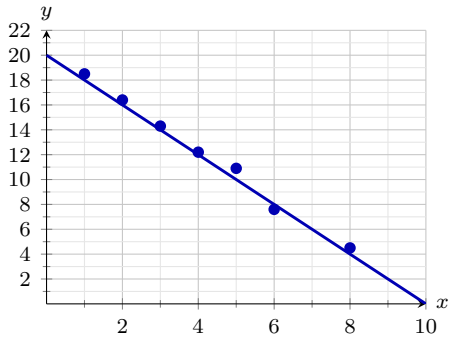
**1.** A town had 8,400 people in 2012 ($t = 0$) and 9,600 in 2020. Assuming linear growth, write $P(t)$, state the rate of change in context, and predict the population in 2030.

**2.** The volume of water in a tank is $V(t) = 560 - 35t$ gallons after $t$ minutes. Interpret the slope and the intercept, and find when the tank is empty.

**3.** Use the regression feature of a calculator on the data. Give the line of best fit and correlation coefficient $r$ to the nearest tenth and hundredth, and describe the correlation.

| $x$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $y$ | 14 | 19 | 20 | 28 | 29 | 35 |

**4.** A car's value $y$ (in thousands of dollars) is shown $x$ years after purchase, with the line $y = -2x + 20$. Describe the correlation, interpret the slope and $y$-intercept, and predict the value at 7 years.




**5.** A line of best fit is $\hat{y} = 5.7x + 45.9$. For the data point $(4, 69)$, find the predicted value and the residual, and say what the residual means.

**6.** Across 40 students, $r = -0.92$ between hours of screen time and hours of sleep. Describe the relationship, find $r^2$, and decide whether this proves screen time reduces sleep.

**7.** A hiker starts at 1,000 ft and climbs 300 ft per hour for 2 hours, then descends 200 ft per hour. Write piecewise $E(t)$ for $0 \le t \le 8$ and find when she is at 1,200 ft on the way down.

**8.** Plant A is measured as shown. Plant B grows by $h = 1.8d + 3$ (cm after $d$ days). Which plant grows faster, and by how much?

| Day | 2 | 6 | 10 |
|---|---|---|---|
| Height A (cm) | 4 | 12.4 | 20.8 |

**9.** Water freezes at 0°C (32°F) and boils at 100°C (212°F). Write $F$ as a linear function of $C$ and find the temperature where both scales agree.

**10.** The 100-meter record in seconds is modeled by $R(t) = 9.7 - 0.04t$, with $t$ years after 1970 (fit to data for $0 \le t \le 50$). Find $R(250)$ and decide whether it is reasonable.

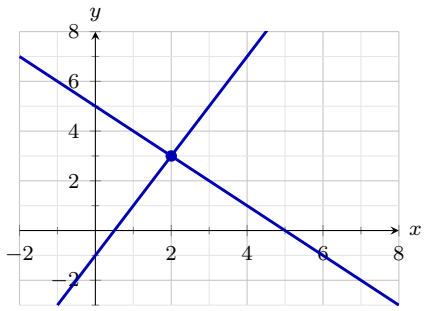
## Algebra 2 Review: Linear Functions & Systems

**Station 3 of 4:** Systems of Equations and Inequalities

Name: _____ Date: _____

*Check every solution in all equations. For inequalities, decide solid or dashed and test a point in each region.*

**1.** The graph shows a system of two lines. Write the system and its solution, then check the solution in both equations.




**2.** Solve by substitution: $y = 2x - 4$ and $3x + 4y = 17$.

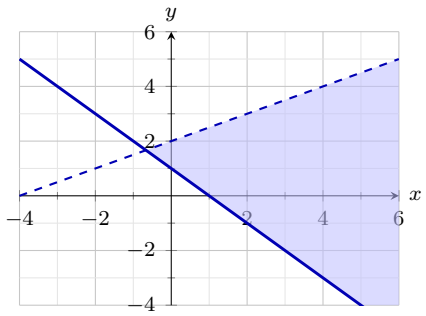
**3.** Solve by elimination: $3x + 2y = 4$ and $5x + 4y = 5$.

**4.** Without graphing, classify each system as one solution, no solution, or infinitely many solutions. (a) $-2x + 6y = 9$, $x - 3y = 4$ (b) $4x - 6y = 10$, $y = \frac{2}{3}x - \frac{5}{3}$.

**5.** A boat goes 24 miles upstream in 3 hours and 24 miles downstream in 2 hours. Write a system and find the boat's speed in still water and the speed of the current.

**6.** A company has costs $C = 1200 + 15n$ and revenue $R = 40n$ for $n$ items. Find the break-even point and say what it means.

**7.** The shaded region is the solution of a system of two linear inequalities (one boundary is dashed). Write the system, then decide whether $(2, 4)$ is a solution.




**8.** Which of $(1, 4)$, $(3, 1)$, $(0, 0)$, $(-2, 5)$ satisfy both $3x - y \geq 2$ and $x + 2y < 6$? Show that $(0, 0)$ fails.

**9.** A bake sale sells cookies for \$2 and brownies for \$3. The class wants at least \$60 and has at most 30 items. Write a system for $c$ cookies and $b$ brownies, and test (15 cookies, 10 brownies) and (10, 10).

**10.** For what value of $k$ does $2x + ky = 5$, $6x + 9y = 8$ have no solution? What would the right side of the second equation need to be for infinitely many?

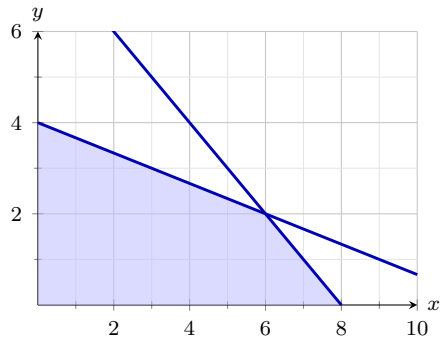
# Algebra 2 Review: Linear Functions & Systems

**Station 4 of 4:** Linear Programming and Three-Variable Systems

Name: _____ Date: _____

*Test every vertex of the feasible region. For three variables, eliminate the same variable twice.*

**1.** The feasible region for $x \geq 0$, $y \geq 0$, $x+y \leq 8$, $x+3y \leq 12$ is shaded. Find its four vertices.




**2.** Use the region from problem 1. Find the maximum of $P = 3x + 5y$ and where it occurs.

**3.** Minimize $C = 4x + 3y$ subject to $x + y \geq 6$, $x + 2y \geq 8$, $x \geq 0$, $y \geq 0$.

**4.** A bakery makes cakes (profit \$12, 2 oven-hours, 1 lb flour) and pies (profit \$9, 1 oven-hour, 3 lb flour). It has 40 oven-hours and 45 lb of flour per day. Write the constraints and find the maximum daily profit and the numbers of each.

**5.** A feasible region has vertices $(1, 1)$, $(1, 5)$, $(4, 4)$, and $(6, 1)$. Find the maximum and minimum of $P = x + 2y$.

**6.** Solve the system: $x + y + z = 2$, $2x - y + 3z = -1$, $x + 2y - z = 7$.

**7.** A jar holds 60 coins (nickels, dimes, quarters) worth \$10.75. There are 5 more dimes than nickels. How many of each coin?

**8.** Classify each system. (a) $x+y+z = 4$, $2x+2y+2z = 9$, $x-y = 1$ (b) $x+y+z = 4$, $2x+2y+2z = 8$, $x - y = 1$. For (b), describe the solutions in terms of $x$.

**9.** Find the quadratic $y = ax^2 + bx + c$ through $(1, 2)$, $(2, 3)$, and $(3, 6)$ by writing a system of three equations.

**10.** Use the region from problem 1 again. Find the maximum of $P = x + 3y$. How many points give that maximum? Name one other than a vertex.

# Algebra 2 Review: Linear Functions & Systems

Answer Key

**Station 1: Equations of Lines, Slope, Parallel and Perpendicular**

**1.** $y = -2x - 1$
$m = \dfrac{-9-5}{4-(-3)} = -2$, *then* $5 = -2(-3) + b$

**2.** $f(x) = 2x + 3$; $f(20) = 43$
*constant rate:* $\dfrac{13-7}{5-2} = 2$

**3.** $y + 2 = -\dfrac{3}{4}(x - 8)$; $y = -\dfrac{3}{4}x + 4$
*point-slope uses* $y - y_1 = m(x - x_1)$, *so* $-(-2)$ *becomes* $+2$

**4.** $5x - 2y = 16$
$y + 3 = \dfrac{5}{2}(x - 2) \Rightarrow 2y + 6 = 5x - 10$

**5.** (a) $y = -\dfrac{2}{5}x + 1$; (b) $y = \dfrac{5}{2}x - \dfrac{27}{2}$
*slope of L is* $-\dfrac{2}{5}$*; perpendicular slope is the negative reciprocal,* $\dfrac{5}{2}$

**6.** $y = -\dfrac{3}{2}x + 3$; $x$-intercept 2
*rise* $-3$ *for run* $2$*;* $y$*-intercept* $3$

**7.** $\ell_1, \ell_2$ parallel (both slope 2); $\ell_1, \ell_3$ perpendicular; $\ell_2, \ell_3$ perpendicular
*slopes are* $2$, $2$, *and* $-\dfrac{1}{2}$

**8.** $C(m) = 28m + 45$; $C(18) = \$549$; $m = 20$ months
$28m + 45 = 605$

**9.** $k = -1$
*need slope* $-3$*:* $\dfrac{-5-4}{2-k} = -3$

**10.** $C(g) = 30$ for $g \le 5$; $C(g) = 30 + 8(g - 5)$ for $g > 5$; $C(9) = \$62$
*4 GB over the limit:* $30 + 8(4)$

**Station 2: Linear Models, Scatter Plots, and Best-Fit Lines**

**1.** $P(t) = 150t + 8400$: grows 150 people per year; $P(18) = 11{,}100$
$\dfrac{9600 - 8400}{8} = 150$

**2.** Slope $-35$: drains 35 gal per minute; intercept 560: starts with 560 gal; empty at $t = 16$ min
$560 - 35t = 0$

**3.** $\hat{y} = 4.1x + 9.9$; $r \approx 0.98$: strong positive correlation
*round only at the end*

**4.** Strong negative correlation; slope $-2$: loses about \$2,000 per year; intercept 20: about \$20,000 new; $y(7) = 6$, about \$6,000
*7 years is inside the data range*

**5.** predicted 68.7; residual $69 - 68.7 = 0.3$: the actual value is 0.3 above the line
*residual = actual − predicted*

**6.** Strong negative linear relationship; $r^2 = 0.8464$, so about 85% of the variation in sleep is explained by the line; correlation does not prove causation

**7.** $E(t) = 1000 + 300t$ for $0 \le t \le 2$; $E(t) = 2000 - 200t$ for $2 < t \le 8$; $t = 4$ hours
*peak is 1,600 ft at* $t = 2$

**8.** A grows 2.1 cm per day, B grows 1.8; A is faster by 0.3 cm per day
$\dfrac{12.4 - 4}{6 - 2} = 2.1$

**9.** $F = \dfrac{9}{5}C + 32$; they agree at $-40°$
*solve* $\dfrac{9}{5}C + 32 = C$

**10.** $R(250) = -0.3$ s: impossible; the model should not be extended far beyond the data

**Station 3: Systems of Equations and Inequalities**

1. $y = -x + 5$ and $y = 2x - 1$; solution $(2, 3)$
   $3 = -2 + 5$ *and* $3 = 2(2) - 1$
2. $(3, 2)$
   $3x + 4(2x - 4) = 17 \Rightarrow 11x = 33$
3. $\left(3, -\frac{5}{2}\right)$
   *double the first equation:* $6x + 4y = 8$
4. (a) no solution (parallel lines, different intercepts); (b) infinitely many (same line)
   *multiply (a)'s second equation by* $-2$*:* $-2x + 6y = -8 \neq 9$
5. boat 10 mph, current 2 mph
   $b - c = 8$ *and* $b + c = 12$
6. $n = 48$ items, where cost and revenue both equal \$1,920; selling more earns a profit
7. $y < \frac{1}{2}x + 2$ and $y \geq -x + 1$; $(2, 4)$: $4 < 3$ is false, so it is not a solution
   *dashed means strict; test a point on each side*
8. only $(3, 1)$; $(0, 0)$ satisfies $x + 2y < 6$ but $3(0) - 0 \geq 2$ is false
   *a point must satisfy every inequality*
9. $2c + 3b \geq 60$, $c + b \leq 30$, $c \geq 0$, $b \geq 0$; $(15, 10)$ works (\$60, 25 items); $(10, 10)$ fails (\$50 < \$60)
10. $k = 3$; with $k = 3$ the second equation must read $6x + 9y = 15$
    *same slope:* $\frac{2}{6} = \frac{k}{9}$*;* $3 \cdot 5 = 15$

**Station 4: Linear Programming and Three-Variable Systems**

1. $(0, 0)$, $(8, 0)$, $(6, 2)$, $(0, 4)$
   *solve* $x + y = 8$ *and* $x + 3y = 12$ *together*
2. maximum 28 at $(6, 2)$
   *values at the vertices:* 0, 24, 28, 20
3. minimum 18 at $(0, 6)$
   *vertices* $(0, 6)$*,* $(4, 2)$*,* $(8, 0)$ *give* 18, 22, 32
4. $2c + p \leq 40$, $c + 3p \leq 45$, $c, p \geq 0$; 15 cakes and 10 pies give \$270
   *vertices* $(0, 0), (20, 0), (0, 15), (15, 10)$ *give \$0, \$240, \$135, \$270*
5. maximum 12 at $(4, 4)$; minimum 3 at $(1, 1)$
   *check all four vertices:* 3, 11, 12, 8
6. $(3, 1, -2)$
   *add eqs. 1 and 3 to get* $2x + 3y = 9$*; eliminate z again from eqs. 1 and 2:* $5x + 2y = 17$
7. 10 nickels, 15 dimes, 35 quarters
   *in cents:* $5n + 10d + 25q = 1075$
8. (a) no solution (the first two planes are parallel); (b) infinitely many: $(x,\ x - 1,\ 5 - 2x)$
   *twice eq. 1 is* 8*, not* 9 *in (a); in (b) the equations agree*
9. $y = x^2 - 2x + 3$
   $a + b + c = 2$*,* $4a + 2b + c = 3$*,* $9a + 3b + c = 6$*; subtract to get* $3a + b = 1$ *and* $5a + b = 3$
10. maximum 12 at $(6, 2)$ and $(0, 4)$ and at every point between them, such as $(3, 3)$ (infinitely many)
    *the objective line is parallel to the edge* $x + 3y = 12$